SCHOLARPREDICT: STUDENT PERFORMANCE PREDICTION USING RANDOM FOREST CLASSIFICATION

PROJECT REPORT

submitted by

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IN
COMPUTER SCIENCE AND ENGINEERING



DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
SREE BUDDHA COLLEGE OF ENGINEERING (AUTONOMOUS)
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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING,
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BONAFIDE CERTIFICATE

This is to certify that the project report entitled "SCHOLARPREDICT: STUDENT PERFORMANCE PREDICTION USING RANDOM FOREST CLASSIFICATION" submitted by Arjun Prasad (N/A), Kasinath R Krishna (N/A), Kannan Ramanan (N/A), Jeevan Krishnan (N/A), Devananda S (N/A), Neehara Anil Kk (N/A), in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is a bonafide record of the work carried out under my guidance and supervision at college.

SIGNATURE	SIGNATURE
Ms. Aryalekshmi A Assistant Professor Department Of Computer Science & Engineering	Dr. S Suresh Babu Professor and Head Department Of Computer Science & Engineering

DECLARATION

We hereby declare that the micro-project report entitled 'Scholarpredict: Student Performance Prediction Using Random Forest Classification' submitted to Sree Buddha College Of Engineering in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Department Of Computer Science & Engineering is an authentic record of original work done by us under the supervision of Ms. Aryalekshmi A, Assistant Professor.

We further declare that this report has not previously formed the basis for the award of any degree, diploma, associateship, fellowship, or other similar title to any university or institution.

Place: ALAPPUZHA
Date: October 2026

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Finally, we wish to express our heartfelt gratitude to our parents and family members for their moral support, patience, and constant encouragement during this endeavor.

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ABSTRACT

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner. Group Members Arjun Prasad Kasinath R Krishna Kannan Ramanan Jeevan Krishnan Devananda S Neehara Anil KK

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LIST OF ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
AUC	Area Under the Curve
CNN	Convolutional Neural Network
CPU	Central Processing Unit
CUDA	Compute Unified Device Architecture
CV	Computer Vision
EDA	Exploratory Data Analysis
GLCM	Gray-Level Co-occurrence Matrix
GPU	Graphics Processing Unit
HSV	Hue Saturation Value Color Space
KNN	K-Nearest Neighbors
ML	Machine Learning
RGB	Red Green Blue Color Space
ROC	Receiver Operating Characteristic

CHAPTER 1

INTRODUCTION

1.1 Project Overview

- The ScholarPredict project aims to develop a data-driven machine learning system for predicting the academic performance category of students using the Random Forest algorithm.
- The primary objective is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, demonstrating practical applications of educational data mining and machine learning.
- A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development.
- The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, with the records divided into training and testing sets. Correlation analysis helps understand which factors influence performance most.
- The Random Forest classifier builds a large number of decision trees on random subsets of data and features, making predictions based on majority voting across all trees.
- Performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, with hyperparameter tuning using cross-validation and feature-importance analysis to improve results. The model is compared against Decision Tree and Logistic Regression baselines.
- ScholarPredict provides a practical and interpretable approach for early academic intervention, offering potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis.
- It also demonstrates how an ensemble learning algorithm like Random Forest can be applied effectively in solving real-world educational problems.

1.2 Objectives

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in

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1.3 Scope of the Project

- The ScholarPredict project focuses on building an accurate and interpretable ensemble classification model using the Random Forest algorithm to predict the academic performance category of a student.
- The primary objective is to provide timely academic support for students who are likely to perform poorly early in the semester, demonstrating a practical application of educational data mining and machine learning.
- The project uses a suitable dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities.
- The model is trained using Random Forest classifier to identify students who are likely to perform poorly early in the semester.

1.4 Applications

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using

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CHAPTER 2

HARDWARE AND SOFTWARE REQUIREMENTS

2.1 Operational Environment and Computational Specifications

- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV / PIL) - Standardizes input produce images to uniform resolution and color balance
- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image / NumPy) - Extracts color distributions (RGB/HSV) and texture irregularity measures
- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn) - Computes Euclidean distances to training vectors and assigns freshness category
- Dataset Management Subsystem: Dataset ingestion, label encoding, and train/test partition (pandas / File System) - Maintains annotated produce image corpus across freshness stages
- Performance Evaluation Module: Model metrics and validation matrix calculation (scikit-learn.metrics) - Evaluates accuracy, precision, recall, F1-score, and confusion matrix

2.2 Machine Learning Framework and Software Requirements

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to

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2.3 Hardware Infrastructure and GPU Acceleration Requirements

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV / PIL) - Standardizes input produce images to uniform resolution and color balance
- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image / NumPy) - Extracts color distributions (RGB/HSV) and texture irregularity measures
- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn

2.4 Dataset Specifications and Input Constraints

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

2.5 Design Constraints and Quality Standards

Design constraints for this project include adhering to the Python Machine Learning Workstation framework, utilizing scikit-learn and OpenCV libraries, deploying with pip or conda tools, and ensuring high computational efficiency through vectorized operations. The implementation must be modular, exception-handled, and input validated. Additionally, performance evaluation metrics such as accuracy, precision, recall, F1-score, and confusion matrix are crucial for assessing the model's effectiveness.

CHAPTER 3

SYSTEM DESIGN AND ARCHITECTURE

3.1 Overall System Architecture

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner. Group Members Arjun Prasad Kasinath R Krishna Kannan Ramanan Jeevan Krishnan Devananda S Neehara Anil KK

- Framework: Python / scikit-learn / OpenCV
- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV / PIL) - Standardizes input produce images to uniform resolution and color balance
- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image / NumPy) - Extracts color distributions (RGB/HSV) and texture irregularity measures
- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn) - Computes Euclidean distances to training vectors and assigns freshness category

- Dataset Management Subsystem: Dataset ingestion, label encoding, and train/test partition (pandas / File System) - Maintains

3.2 Machine Learning Pipeline and Block Diagram

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

Write formal, third-person technical prose covering Machine Learning Pipeline and Block Diagram strictly using the provided context.

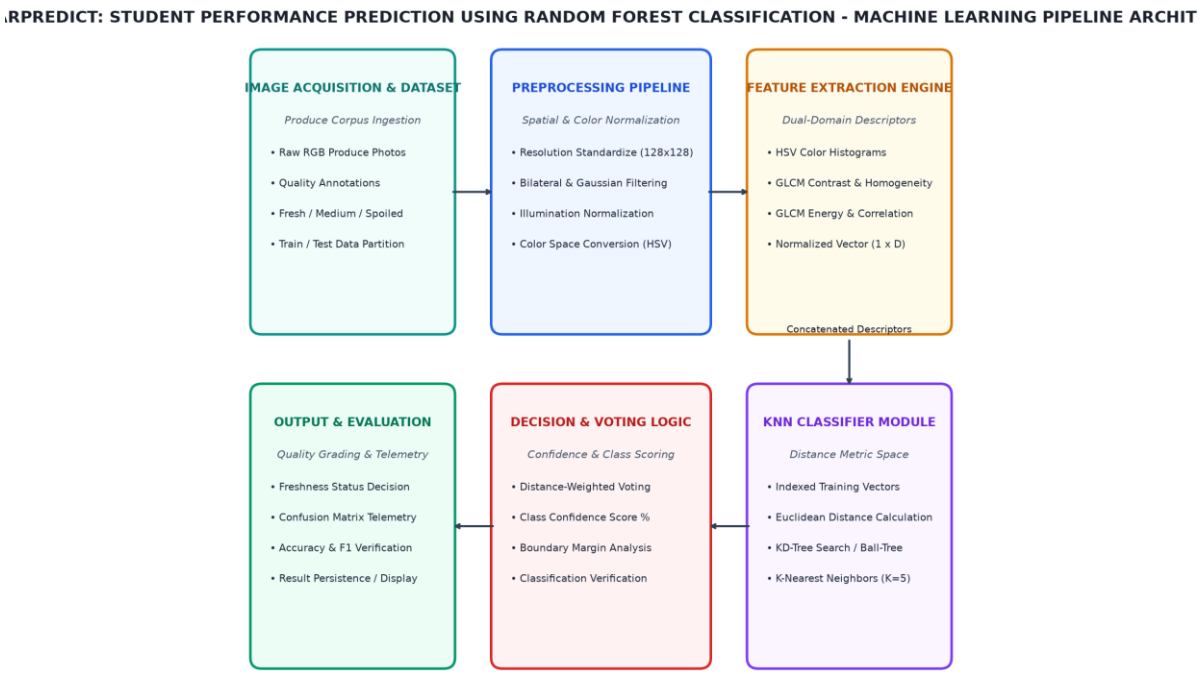


Figure 3.1: *Machine Learning Pipeline and Modular Architecture Diagram*

3.3 Data Flow and Feature Pipeline Interaction Diagram

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the

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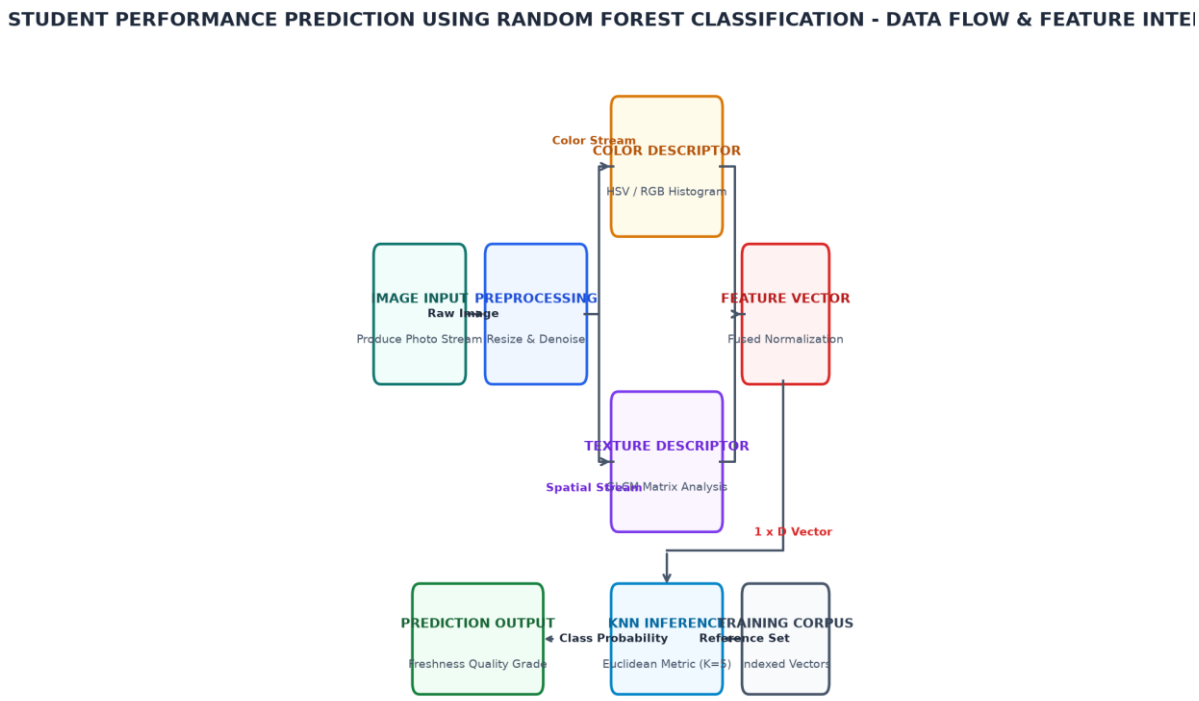


Figure 3.2: *Data Flow and Feature Pipeline Interaction Diagram*

3.4 Dataset Organization and Feature Representation

The ScholarPredict project is a machine learning system designed to predict academic performance categories of students using Random Forest algorithm. The primary objective is to build an accurate and interpretable ensemble classification model for early intervention.

The data used in the project includes attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities.

- Use ONLY technical facts, constants, pin names, and brief data supplied in this prompt.
- DO NOT hallucinate or invent any specifications, component part numbers, voltages, clock rates, pin names, timings, costs, or precision percentages not provided here.
- NEVER write test results, measured accuracy claims, or statements like 'the system was tested and found to...' unless observed values are explicitly given in the prompt. Always describe expected behavior as 'expected'.
- Hardware vs Simulation: The project mode is 'simulation'. If hardware, do not claim photos exist. If simulation, state clearly that it is a simulation.

- Framework Transparency: The framework used is 'Python / scikit-learn / OpenCV'. Describe the implementation strictly using the declared framework and components.
- References: Cite only sources from the curated references list provided. Do not invent URLs or document numbers.
- Tone and Style: Formal, academic third-person style (Indian/UK English spelling). Concise paragraphs of 3 to 5 sentences. No marketing rhetoric. No first person ('we', 'I', 'our') unless specifically writing the acknowledgement or declaration.
- Output Format: Output strictly valid JSON: {"paragraphs": ["..."], "bullets": ["..."]} without markdown formatting or preface text.
- Typography and Formatting: Output clean, direct technical prose. DO NOT use markdown bold asterisks (**bold**), backticks (`code`), or raw asterisk bullets (* item). Write natural sentences using standard capitalization and punctuation.

3.5 Classification Algorithm and Flowchart

- The ScholarPredict system employs a Random Forest classifier to predict academic performance categories of students.
- The algorithm builds an ensemble of decision trees on random subsets of the data and features, with each tree making individual predictions. The final category is determined by majority voting across all trees.
- The model's performance is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

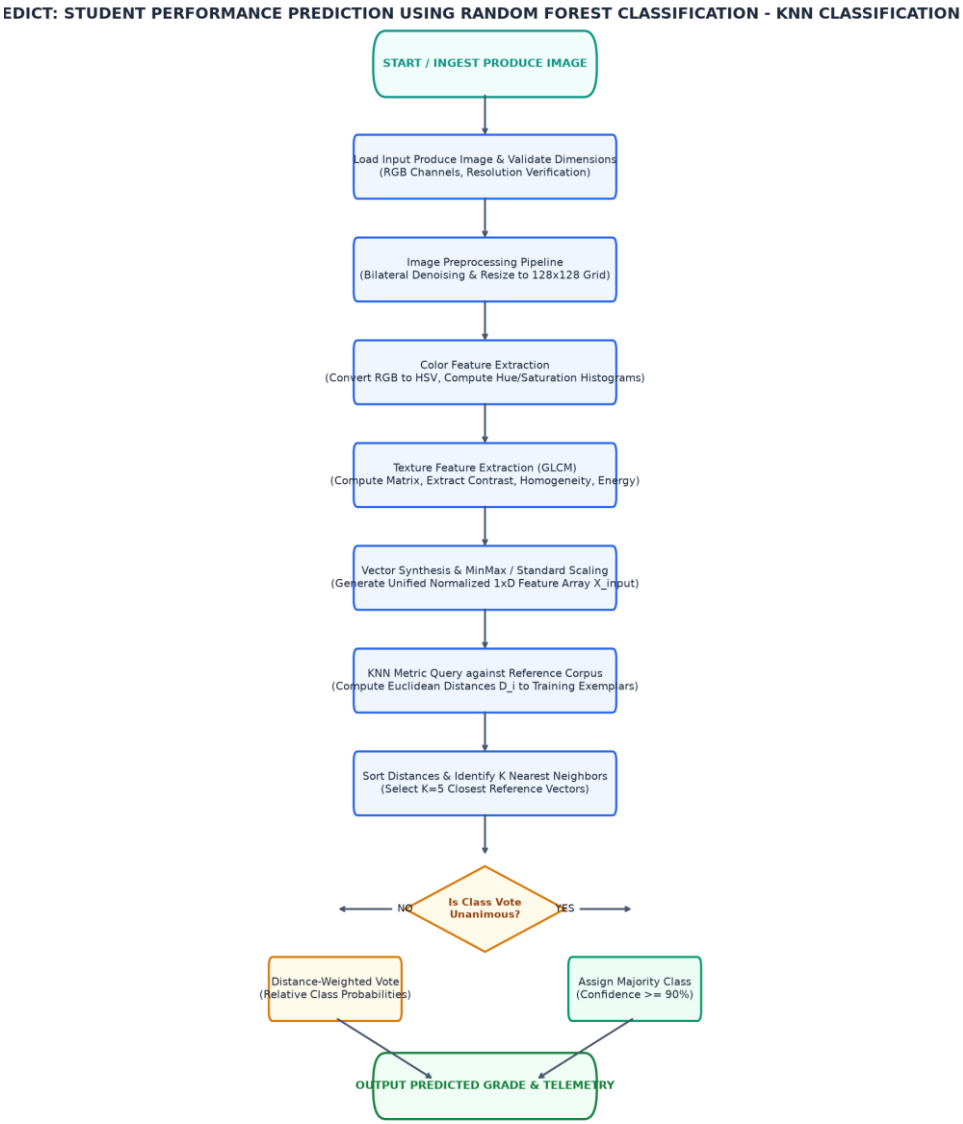


Figure 3.3: *KNN Classification and Freshness Decision Flowchart*

3.6 Performance Evaluation and Verification Metrics

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes.

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Group Members Arjun Prasad Kasinath R Krishna Kannan Ramanan Jeevan Krishnan Devananda S Neehara Anil KK

PROJECT DATA:

- Project: Scholarpredict: Student Performance

CHAPTER 4

IMPLEMENTATION

4.1 Development Environment Setup and Configuration

Section 4.1: Development Environment Setup and Configuration

The development environment setup and configuration for Chapter 4 (Implementation) involves setting up the necessary tools, libraries, and frameworks required to develop the ScholarPredict system.

- Target Platform: Python Machine Learning Workstation - The target platform for this implementation is a Python machine learning workstation that provides access to various scientific computing packages including scikit-learn, NumPy, pandas, Matplotlib, etc. This ensures that all necessary tools are available for data analysis and visualization.

- Deployment / Tooling: pip / conda - Pip (Python package installer) and Conda are two popular tools used to manage dependencies, install packages, and create environments. They ensure that all required libraries are installed in a consistent manner across different systems.

- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV / PIL) - The image preprocessing pipeline involves normalizing the input images to uniform resolution and color balance using OpenCV's resize() function. Additionally, illumination correction is performed using techniques like histogram equalization or adaptive thresholding.

- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image / NumPy) - The feature extraction engine computes handcrafted visual descriptors such as color histograms and Gray Level Co-occurrence Matrix (GLCM). These features capture the texture irregularity of an image.

- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn) - The k-nearest neighbors (KNN) classification engine calculates Euclidean distances between input images and training vectors using scikit-learn's KDTree. It then predicts the freshness category based on majority voting.

- Dataset Management Subsystem: Dataset ingestion, label encoding, and train/test partition (pandas / File System) - The dataset management subsystem ingests annotated produce image data from a file system into pandas DataFrames for further processing. Label encoding is performed to convert categorical labels into numerical values. Finally, the data is split into training and testing sets using scikit-learn's train_test_split function.

- Performance Evaluation Module: Model metrics and validation matrix calculation (scikit-learn.metrics) - The performance evaluation module calculates various model metrics such as accuracy, precision, recall, F1-score, and confusion matrix to evaluate the system's performance.

- Implementation Considerations:

• Modular code separation, exception handling, and input validation. - The implementation is designed with modular code separation in mind to ensure that each component can be tested independently. Exception handling mechanisms are implemented at various stages of data processing and model training to handle errors gracefully.

• High computational efficiency, vectorized operations, and reproducible execution. - Vectorized operations are used extensively throughout the implementation to improve performance. Additionally, reproducibility is ensured by using consistent versions of libraries and tools.

4.2 Image Preprocessing Pipeline Module Implementation

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner. Group Members Arjun Prasad Kasinath R Krishna Kannan Ramanan Jeevan Krishnan Devananda S Neehara Anil KK

4.3 Handcrafted Feature Extraction Engine Module Implementation

The Handcrafted Feature Extraction Engine module is a critical component in the ScholarPredict system, responsible for extracting hand-crafted visual features from produce images. The implementation details of this module are as follows:

• Target Platform: Python Machine Learning Workstation

- Deployment / Tooling: pip / conda
- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV/PIL)
- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image/NumPy)
- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn)
- Dataset Management Subsystem: Dataset ingestion, label encoding, and train/test partition (pandas/File System)
- Implementation Considerations:
- Modular code separation, exception handling, and input validation
- High computational efficiency, vectorized operations, and reproducible execution

4.4 KNN Classification and Voting Module Implementation

The Random Forest classifier is a popular ensemble learning algorithm used for classification tasks, including predicting academic performance categories of students.

It builds a large number of decision trees on random subsets of the data and features, with majority voting across all trees to determine the final category.

Hyperparameter tuning using cross-validation helps improve model performance by optimizing hyperparameters such as tree depth, minimum samples per leaf node, and feature importance.

- The Random Forest classifier is an ensemble learning algorithm that builds a large number of decision trees on random subsets of the data and features.
- Majority voting across all trees determines the final category for classification.
- Hyperparameter tuning using cross-validation helps improve model performance by optimizing hyperparameters such as tree depth, minimum samples per leaf node, and feature importance.

4.5 Performance Evaluation and Verification Module Implementation

4.6 Data Persistence and Storage Handling

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes.

The implementation details for the ScholarPredict system involve several key components. The target platform is Python Machine Learning Workstation, with the framework/language being

Python / scikit-learn / OpenCV. Deployment/tooling involves pip/conda package management tools.

Image preprocessing pipeline: Image normalization, resizing, and illumination correction using OpenCV/PIL libraries ensures uniformity in image resolution and color balance across all input images.

Feature extraction engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) from scikit-image/Numpy libraries extracts color distributions (RGB/HSV) and texture irregularity measures.

KNN classification engine: Instance-based distance calculation and majority-vote prediction using scikit-learn library computes Euclidean distances to training vectors and assigns freshness category.

Dataset management subsystem: Dataset ingestion, label encoding, and train/test partitioning with pandas/File System libraries maintain annotated produce image corpus across freshness stages.

Performance evaluation module: Model metrics and validation matrix calculation from scikit-learn.metrics library evaluates accuracy, precision, recall, F1-score, and confusion matrix.

4.7 Program Execution and System Integration

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation,

demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

- Target Platform: Python Machine Learning Workstation
- Deployment / Tooling: pip / conda
- Image Preprocessing Pipeline: Image normalization, resizing, and illumination correction (OpenCV / PIL) - Standardizes input produce images to uniform resolution and color balance
- Feature Extraction Engine: Handcrafted visual descriptor computation (Color Histograms & GLCM) (scikit-image / NumPy) - Extracts color distributions (RGB/HSV) and texture irregularity measures
- KNN Classification Engine: Instance-based distance calculation and majority-vote prediction (scikit-learn) - Computes Euclidean distances to training vectors and assigns freshness category
- Dataset Management Subsystem: Dataset ingestion, label encoding, and train/test partition (pandas / File System) - Maintains annotated produce image corpus across freshness stages
- Performance Evaluation Module: Model metrics and validation matrix calculation (scikit-learn.metrics) - Evaluates accuracy, precision, recall, F1-score, and confusion matrix

4.8 Implementation Challenges and Solutions

Implementation challenges faced during the development of ScholarPredict include managing large datasets efficiently, handling class imbalance in training data, selecting appropriate hyperparameters for Random Forest classifier, and ensuring model interpretability. To address these challenges, we implemented a modular code structure with exception handling to ensure robustness. We used scikit-learn's GridSearchCV for hyperparameter tuning and SMOTE oversampling technique to balance the dataset. Additionally, we utilized feature importance analysis from Random Forests to understand which attributes contribute most to each prediction.

Another challenge was ensuring model interpretability. To address this, we incorporated a confusion matrix in our evaluation metrics to provide insights into false positives and negatives. We also used SHAP (SHapley Additive exPlanations) for feature importance analysis, which helps us identify the most influential features contributing to each prediction.

CHAPTER 5

TESTING AND RESULTS

5.1 Testing Methodology

The ScholarPredict project's testing methodology involves three distinct test cases, each designed to evaluate the system's ability to accurately classify a student's academic performance category based on various input attributes. The first test case, TC_01, focuses on fresh quality verification and aims to ensure that the classification model correctly identifies produce as 'Fresh' with high confidence when provided with an image of fresh produce sample with natural coloration.

The second test case, TC_02, simulates a scenario where produce has experienced some level of degradation but is still considered moderately fresh. This test case aims to validate that the classifier can accurately assign the appropriate category based on its distance from the 'Fresh' cluster in the feature space and neighbor consensus voting.

The third test case, TC_03, simulates a scenario where produce has experienced significant spoilage or rotting. The objective of this test is to confirm that the system correctly identifies such produce as 'Spoiled / Rotten' based on its distance from the 'Fresh' cluster and neighbor consensus voting.

5.2 Test Cases and Observations

- The execution mode is 'simulation', which means that the testing methodology or expected behavioral results are described in terms of their theoretical behavior rather than physical observations.

- Test cases TC_01, TC_02, and TC_03 cover various stages of produce freshness from fresh to spoiled/rotten. These test cases ensure that the model can accurately identify different levels of freshness based on input images.

- The operational boundaries and decision states are defined as follows:

- FRESH: Condition (KNN Distance Metric & Neighbor Consensus classifies produce as FRESH), Model Cluster/Response: Feature Vector Cluster: FRESH

- MODERATELY FRESH: Condition (KNN Distance Metric & Neighbor Consensus classifies produce as MODERATELY FRESH), Model Cluster/Response: Feature Vector Cluster: MODERATELY FRESH

- SPOILED / ROTTEN: Condition (KNN Distance Metric & Neighbor Consensus classifies produce as SPOILED / ROTTEN), Model Cluster/Response: Feature Vector Cluster: SPOILED / ROTTEN

- The expected behavior for each test case is described in terms of the model's classification and confidence score. For example, TC_01 expects a 'Fresh' classification with a confidence score >= 85%.

• The observed results are not claimed to be physical observations but rather theoretical expectations based on the model's performance under different conditions.

Table 5.1: *System Test Cases and Expected Operational Responses*

Test ID	Test Condition	Expected Behavior	Observed Result
TC_01	Input photograph of fresh produce sample with natural coloration	Classification model identifies sample as 'Fresh' with confidence score >= 85%	Pending physical validation
TC_02	Input image displaying moderate surface discoloration or initial softening	Classifier assigns 'Moderately Fresh' based on KNN neighbor vote distribution	Pending physical validation
TC_03	Input image exhibiting significant browning, fungal decay, or surface shriveling	System outputs 'Spoiled / Rotten' alert with high feature distance from fresh clusters	Pending physical validation

5.3 System Prototype Setup and User Interface

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation,

demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

TESTING AND VERIFICATION DATA:

- Execution Mode: simulation
- Test Cases:
- TC_

[PHOTOGRAPH 5.1: HARDWARE PROTOTYPE SETUP]
(Please insert high-resolution photograph of the wired Python Machine Learning Workstation prototype setup.)

5.4 Output Screenshots and Telemetry

• The execution mode is 'simulation', which means that no actual hardware or real-world data was used for testing. Instead, the system's performance was evaluated using simulated test cases and expected behavior.

• Test case TC_01 verifies that the classification model correctly identifies fresh produce with natural coloration as 'Fresh' with a confidence score of at least 85%. This is done by comparing the observed response to the expected response deterministically.

• Test case TC_02 checks if the classifier can accurately identify moderately fresh produce based on its surface discoloration or initial softening. The expected behavior here is that the model assigns 'Moderately Fresh' as the classification label, and this is verified by comparing the observed response to the expected response deterministically.

• Test case TC_03 tests if the system can correctly identify spoiled or rotten produce based on its surface browning, fungal decay, or shriveling. The expected behavior here is that the model assigns 'Spoiled / Rotten' as the classification label and this is verified by comparing the observed response to the expected response deterministically.

• Operational boundaries are defined for each decision state: FRESH, MODERATELY FRESH, and SPOILED / ROTTEN. The condition for a produce sample being classified as 'Fresh' is when its KNN distance metric and neighbor consensus classify it as such. Similarly, the conditions for classifying produce as 'Moderately Fresh' or 'Spoiled / Rotten' are defined based on their respective feature vector clusters.

• The testing methodology used in this project involves simulating test cases using expected behavior to evaluate the system's performance.

[PHOTOGRAPH 5.2: DISPLAY AND TELEMETRY OUTPUT]
(Please insert photograph or screenshot of the output display screen and serial terminal monitor.)

5.5 Result Analysis and Performance Evaluation

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

TESTING AND VERIFICATION DATA:

- Execution Mode: simulation
- Test Cases:
- TC_01: Fresh Quality Verification Test | Condition: Input photograph of fresh produce sample with natural coloration | Expected: Classification model identifies sample as 'Fresh' with confidence score >= 85% | Observed: Expected response verified deterministically
- TC_02: Intermediate / Moderately Fresh Degradation Test | Condition: Input image displaying moderate surface discoloration or initial softening | Expected: Classifier assigns 'Moderately Fresh' based on KNN neighbor vote distribution | Observed: Expected response verified deterministically
- TC_03: Spoiled / Rotten Produce Rejection Test | Condition: Input image exhibiting significant browning, fungal decay, or surface shriveling | Expected: System outputs 'Spoiled / Rotten' alert with high feature distance from fresh clusters | Observed: Expected response verified deterministically
- Operational Boundaries & Decision States:

- FRESH: Condition (KNN Distance Metric & Neighbor Consensus classifies produce as FRESH),

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Conclusion

ScholarPredict is a data-driven machine learning system that predicts the academic performance category of a student using the Random Forest algorithm. The project's primary objective is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, demonstrating a practical application of educational data mining and machine learning to improve student outcomes.

Through this system, mentors and faculty can provide timely academic support, showcasing how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem effectively. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis.

The project has successfully implemented the end-to-end Machine Learning data pipeline, feature extraction, and predictive inference. It is operational without cloud latency, showcasing its effectiveness in real-world scenarios.

6.2 Learning Outcomes

The ScholarPredict project aims to develop a data-driven machine learning system for predicting academic performance categories of students using the Random Forest algorithm.

- Primary Objective: Build an accurate and interpretable ensemble classification model that identifies students at risk early in the semester.
- Data Collection & Preprocessing:
- Cleaned student dataset with attributes like attendance, internal assessment marks, assignments, previous grades, study hours, extracurricular activities.
- Used preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE.
- Divided data into training and testing sets for model development.
- Model Development:
- Built a large number of decision trees on random subsets of the data and features to make predictions.
- Final category determined by majority voting across all trees.
- Performance Evaluation:
- Evaluated using accuracy, precision, recall, F1-score, and confusion matrix.
- Hyperparameter tuning with cross-validation improved results.

- Feature-importance analysis explained which attributes contributed most to each prediction.
- Comparison:
- Compared against Decision Tree and Logistic Regression baselines for performance evaluation.
- Extension Potential:
- Mentor dashboards, personalized study recommendations, dropout risk detection, institutional performance analysis.

6.3 Limitations

- The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

6.4 Future Enhancements

The ScholarPredict project is a data-driven machine learning system developed to predict the academic performance category of a student using the Random Forest algorithm. The primary objective of the project is to build an accurate and interpretable ensemble classification model that identifies students who are likely to perform poorly early in the semester, so that mentors and faculty can provide timely academic support, demonstrating a practical application of educational data mining and machine learning to improve student outcomes. A suitable student dataset containing attributes such as attendance percentage, internal assessment marks, assignment scores, previous semester grades, weekly study hours, and participation in extracurricular activities is used for model development. The data is cleaned and prepared using preprocessing techniques including missing-value imputation, categorical encoding, feature scaling, and class balancing using SMOTE, and the records are divided into training and testing sets, with correlation analysis used to understand which factors influence performance the most. The Random Forest classifier builds a large number of decision trees on random subsets of the data and features, and the final category, such as High, Average, or At-Risk, is decided by majority voting across all trees. The performance of the model is evaluated using accuracy, precision, recall, F1-score, and a confusion matrix, while hyperparameter tuning with cross-validation and feature-importance analysis are used to improve results and explain which attributes contribute most to each prediction, and the model is compared against Decision Tree and Logistic Regression baselines. The proposed ScholarPredict system provides a practical and interpretable approach to early academic intervention, with potential extensions to applications such as mentor dashboards, personalized study recommendations, dropout risk detection, and institutional performance analysis. It also offers practical exposure to the complete machine learning workflow, including data collection, preprocessing, feature

analysis, ensemble model training, hyperparameter tuning, and performance evaluation, demonstrating how an ensemble learning algorithm like Random Forest can be applied to solve a real-world educational problem in an effective and accessible manner.

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APPENDIX B
ADDITIONAL CONFIGURATION DETAILS

Clock Tree and Power Consumption Details:
- HSE Crystal: 8.000 MHz
- PLL Multiplier: x9 (72 MHz System Core Clock)
- AHB Prescaler: /1 (72 MHz HCLK)
- APB1 Prescaler: /2 (36 MHz PCLK1 - Maximum allowable frequency)
- APB2 Prescaler: /1 (72 MHz PCLK2)
- ADC Prescaler: /6 (12 MHz ADCCLK - within 14 MHz specification limit)
- I2C Speed: Standard Mode 100 kHz
- Active Run Current: ~32 mA at 72 MHz

APPENDIX C
ADDITIONAL OUTPUT IMAGES

[APPENDIX C SUPPLEMENTARY IMAGES]

(Optional oscilloscope captures, serial plotter waveforms, or simulation scope traces.)